Problem A. Aesthetics in poetry

Time Limit 1000 ms
Mem Limit 262144 kB
OS Windows

One of the greatest poets in Portuguese language was Luiz Vaz de Camões, born in Lisbon circa 1524. Camões works are fundamental reading for high school students in countries such as Brazil or Portugal. Many consider his 5- and 7-syllable poems and his sonnets (with 14 verses) to be the most important lyrical work in Portuguese of all time.

Camões was a visionary in his own time. His work has been studied from many aspects: historical, cultural, symbolical, and so on. Carlitos "the efficient" Nunes is studying aesthetics in Camões' poetry. He is especially interested in a metric based on the size of the verses of a poem. We define this metric as follows. Suppose we have a poem made of N verses. We say that the poem is K-elegant if $K > 1$, N is a multiple of K and, moreover, there are exactly N/K verses whose sizes, when divided by K, have remainders equal to i, for $i = 0, 1, \ldots, K - 1$. Note that a single poem may be K-elegant for several distinct values of K.

Carlitos already processed the data from poems by Camões and now he needs your help to determine the smallest elegance in each of the poems.

Input

The input has two lines. The first line has an integer N, the number of verses in the poem. In the second line you are given N integers, say $\ell_1, \ell_2, \ldots, \ell_N$, such that ℓ_i represents the size of the ith verse of the poem.

Constraints

- $1 \le N \le 2 \cdot 10^3$
- $1 \le \ell_i \le 10^9$

Output

A single integer K, the smallest integer such that the poem is K-elegant. If no such integer exists, print "-1".

Examples

Input	Output
6 3 6 1 7 8 14	2

Input	Output
5 17 21 14 35 13	5

Note

In the first case, if $K = 2$, then 6, 8 and 14 (respectively, 3, 1 and 7) have remainder 0 (respectively, remainder 1) when divided by K.

Problem B. Maximizing Advertising

Time Limit 1000 ms
Mem Limit 262144 kB
OS Windows

In Portugal there are very popular political parties, the Social Democrat Party (PSD, in Portuguese) and the Socialist Party (PS, in Portuguese). There will be presidential elections in Portugal in 2019 and the political parties are working on the best way to run advertising campaigns to reach the maximum number of voters.

Due to the high cost of elections advertising, PSD and PS wish to broker a deal and split the costs of advertising. The people in charge of the negotiation have data about political preference of some people in Portugal. They have been processing this information and have thus obtained the geographical position of people that are likely voters for PSD and PS. This position is represented by a point in the plane. To calculate the expenses, they wish to find two disjoint areas that, when summing likely voters for these parties, the result is the largest possible. To facilitate the computations, they decided to consider both areas to be rectangles with sides parallel to the axes. In other words, they wish to find two rectangles, one representing PSD and the other representing PS, such that the sum of the number of likely voters of PSD in the first rectangle and the number of the likely voters of PS in the second rectangle is maximum. You were hired to develop a software that fulfills this task.

Input

The first line has an integer N, the number of people whose data will be considered. The next N lines describe each person's data. The ith of these N lines has two integers and a character, say x_i, y_i and p_i. The integers represent a point (x_i, y_i) and p_i represents the political preference of the ith person.

Constraints

- $1 \leq N \leq 10^6$
- $-10^9 \leq x_i, y_i \leq 10^9$
- $p_i =$ "b" or $p_i =$ "w"

Output

The maximum value that can be obtained when summing the number of likely voters in each rectangle. In this problem, we consider that points in the boundary are part of the rectangle.

Examples

Input	Output
3 0 0 b 0 1 b 1 1 b	3

Input	Output
4 0 0 b 1 1 w 2 2 b 3 3 w	3

Input	Output
4 -1 -1 b 0 0 b 1 1 w 2 2 w	4

Problem C. Group work

Time Limit 1000 ms
Mem Limit 262144 kB
OS Windows

Recent studies in Education show great pedagogical benefits of working in groups, especially in the freshman and sophomore years in high school. Different groups of various sizes are great and allow students to understand their differences and to train in teamwork. A teacher at a modern school wants to know how many different groups he can form in this class of N students. The only requirement is that a group must have at least 2 students, and two groups are distinct if at least one student is in one of them and no in the other.

Your task in this problem is, given N, compute the number of distinct groups that is possible to create in the class.

Input

A single line with an integer N.

Constraints

- $1 \leq N \leq 30$

Output

A single line with the number of distinct groups that is possible to have in the class.

Examples

Input	Output
2	1

Input	Output
3	4

Note

In the first example we have two students, therefore we can only have one group. In the second example we have three students, say A, B and C. We can form the groups AB, AC, BC and ABC.

Problem D. Running a penitentiary

Time Limit 2000 ms
Mem Limit 262144 kB
OS Windows

The penitentiary at Viana do Castelo is one of the largest ones in Europe. The building has thousands of jail cells, laid out linearly in several levels, and numbered so that adjacent numbers are given to adjacent cells, be it adjacent cells on the same level, or one on top of the other, or connected by an adjacent stairway. It is not really known why, but the jail cells are numbered starting from a negative number, so that cell number zero is right in the middle of the penitentiary, exactly at the center of the middle level of the building. Probably the first warden, who assigned the cells numbering, did this for entertainment . . .

Each guard in the penitentiary gets an interval of cells to watch. He must go over these cells and check that the inmates did not escape, are doing okay, and so on. The warden does two operations depending on the intervals of cells assigned to each guard:

1. Given a guard i, change the interval of cells assigned to him to $[\ell, r]$;
2. Given an interval of guards $[a, b]$, find how many cells are watched by all the guards in this interval at the moment.

Your task in this problem is to read the number N of guards in the penitentiary and, for each guard, the interval of cells he is in charge of watching over. Next, you will read the Q operations carried out by the warden, by executing an operation of the first type or by answering the query for operations of the second type.

Input

The first line has two integers N and Q, the number of guards and the number of operations to carry out, respectively. The next N lines describe the guards information. The ith of these lines has two integers, L_i and R_i, the interval of cells assigned to the ith guard. Finally, the next Q lines describe the operations chosen by the warden. Each one of these lines represents an operation as follows

- "C i ℓ r": this indicates changing the interval of cells of the ith guard to $L_i = \ell$ and $R_i = r$.
- "? a b" : this indicates the query, given the interval of guards $[a, b]$, how many cells are watch over by all the guards in this interval.

Note that, at each moment, each guard has a single interval of cells assigned to him.

Constraints

- $1 \le N, Q \le 2 \cdot 10^5$
- $-10^9 \le L_i \le R_i \le 10^9$
- For each operation of type `'C'`, $1 \le i \le N$ and $-10^9 \le \ell \le r \le 10^9$
- For each operation of type `'?'`, $1 \le a \le b \le N$
- There is at least one operation of type `'?'`

Output

For each operation of type `'?'`, print a line with the answer.

Examples

Input	Output
3 4 1 100 5 10 7 20 ? 1 3 ? 2 3 C 1 8 8 ? 1 3	4 4 1

Input	Output
2 3 1 2 9 10 ? 1 1 ? 2 2 ? 1 2	2 2 0

Problem E. Wine Production

Time Limit 1500 ms
Mem Limit 262144 kB
OS Windows

Porto is known worldwide for its wine production. These wines are produced in the outskirts of the city and then aged in oak barrels to obtain their unique flavor. The wineries must follow strict production standards to get the seal of approval that helps with sales. To attain these very high standards, the temperatures at the vineyards are measured daily and must satisfy very specific requirements, according to studies by experts.

In order for a grape to reach optimum development, it is important to known how many different temperatures it faced during its growth and in how many days each of these temperatures were repeated. Recent studies show that the quality of the wine is deeply correlated with the following number. Consider the growth period for the grape and count, in those days, how many distinct temperatures we had, and for each temperature, how many times it was repeated in the period. We say that the grapevine has quality x if we have at least x distinct temperatures that were repeated in x days during the growth of its grapes.

Your task is to compute the quality of production of each grapevine in the vineyard. You are given the temperatures measured in N days in the vineyard. Next, for each one of the Q grapevines in the vineyard, you are given the start and end days of their growth periods. For each one of these grapevines, you must compute the maximum quality of their production.

Input

The first line has two integers N and Q, the number of days of temperature measurements in the vineyard and the number of grapevines, respectively. The second line has N blank-separated integers. The ith integer is the temperature measured on day i. Finally, the next Q lines describe each one of the grapevines. The ith of these lines has two integers, ℓ_i and r_i, the starting and ending day of growth of the grapes in the ith grapevine.

Constraints

- $1 \leq N, Q \leq 3 \cdot 10^4$
- The temperature each day is between -10^9 and 10^9
- $1 \leq \ell_i \leq r_i \leq N$

Output

Print Q lines, the ith of which is the maximum quality of grapevine i.

Examples

Input	Output
6 3 1 2 3 1 2 1 1 6 2 4 1 5	2 1 2

Note

In the first grapevine we had 3 distinct temperatures. Only one gets repeated at least three times and two of them are repeated twice, so the maximum quality of the grapevine is 2.

In the second grapevine the temperatures were all distinct, so the maximum quality is 1. Finally, the maximum quality of the third grapevine is 2, since we had two temperatures that were repeated twice.

Problem F. A story about tea

Time Limit 4000 ms
Mem Limit 262144 kB
OS Windows

Tea is second most consumed beverage in the world after water, especially in the United Kingdom, where about 84% of the population has tea every day. This British tradition with tea is a well known fact. What few people know is that English tea has its origins in Portugal. In 1662, when Catarina de Bragança married King Charles II of England, she traveled to London and took with her some tea leaves. The story goes that the boxes where these leaves were carried were labeled as "Transport of Aromatic Herbs", later abbreviated in Portuguese to TEA.

The historian Gaia "the Curious" Fontenelle is studying some old documents that describe how tea was transported from Portugal to England. Initially, N boats were anchored in Portugal. Each boat was assigned a distinct integer between 1 and N. The documents say that the tea was to be transported in a total of K trips. These trips would use ports in Portugal, China and England. After many ruminations, Gaia reached these conclusions:

1. In each one of the K trips only one boat would move between two ports. Moreover, this boat was the smallest numbered when considering the boats stationed in **both** ports.
2. The trips would take place one after the other, and there never were two boats traveling at the same time.
3. Boat N would end its itinerary when it arrived in England. For $i < N$, boat i would only end its itinerary when it arrived in England and the itinerary of boat $i + 1$ had also ended.

Gaia quickly noticed that, with these rules, each port looks like a stack, where a trip corresponds to moving the smallest boat (in the top) of the origin stack to the destination stack (while satisfying constraint 1). Gaia was used to dealing with imprecise or downright false sources of information, so she started playing in her notebooks and she realized that, depending on the values of N and K, sometimes it was impossible to move all N boats (even while using the port in China). For this reason, for each report about tea transport, she wants to find out whether it was possible or not.

Gaia got hungry and left for coffee, chocolate and taking pictures of the sunset. You are an intern in her team and wish to impress her (maybe you will get the job this way). Thus, you decided to solve the problem before she returns.

Input

The input has a single input line with integers N and K, the number of boats and the total number of trips by the boats, respectively.

Constraints

- $1 \leq N \leq 21$
- $1 \leq K \leq 2^{21}$

Output

In the first line print "Y" (without quotes) if it is possible to move the N boats from Portugal to England, while respecting the constraints described above. Otherwise, print "N". If the answer is affirmative, print K lines that describe a itinerary of trips that moves the N boats. To represent a trip from port X to port Y, print "X Y". Each port is represented by a character as follows:

- Portugal: "A"
- China: "B"
- England: "C"

If there is more than one possible itinerary, any one will be accepted.

Examples

Input	Output
2 3	Y A B A C B C

Input	Output
2 1	N

Input	Output
2 6	Y A B B A A C C B A C B C

Note

The output of the first example described the following itinerary:

1. Boat 1 travels from Portugal to China.
2. Boat 2 travels from Portugal to England.
3. Boat 1 travels from China to Portugal.

Problem G. Meme Wars

Time Limit 1000 ms
Mem Limit 262144 kB
OS Windows

Altercations between Brazil and Portugal go a long way back. The most recent such skirmish happened in 2016 when these two countries had a "meme war". The journalist Fiorentina "sagacious" Azzellini is studying this event in her work of describing ways in which social networks have changed how people interact.

Brazil flooded the internet with a huge amount of memes, and won the contest in three days. Fiorentina, while investigating how such a huge amount of images could be created, found that some of them could be constructed very simply. The meme was made of an image that had the national flags of Brazil and Portugal, and one word. The word $S(x)$ was created from a letter x and has the following composition:

$$S(x) = \text{'a'}, \text{ if } x = \text{'a'}. \text{ Otherwise, } S(x) = S(p(x)) \cdot x \cdot S(p(x)),$$

where $p(x)$ is the letter that precedes x in the alphabet and $A \cdot B$ denotes the concatenation of words A and B. Fiorentina started playing with distinct letters and wrote in her notebook

S('b') = 'aba',

S('c') = 'abacaba',

...

Sagacious as she is, Fiorentina quickly noticed that the word S('z') is huge. Not willing to write it down in her notebook, but still curious about some of the letters in this word, she asked for you to help. She would like to know what the letter in the nth position of the word S('z') is.

Input

A single integer n.

Constraints

- $1 \le n \le 10^3$

Output

A single character, the letter in the n-th position of the word $S(\texttt{'z'})$.

Examples

Input	Output
1	a

Problem H. Download Speed

Time Limit 1000 ms

Mem Limit 524288 kB

Consider a network consisting of n computers and m connections. Each connection specifies how fast a computer can send data to another computer.

Kotivalo wants to download some data from a server. What is the maximum speed he can do this, using the connections in the network?

Input

The first input line has two integers n and m: the number of computers and connections. The computers are numbered $1, 2, \ldots, n$. Computer 1 is the server and computer n is Kotivalo's computer.

After this, there are m lines describing the connections. Each line has three integers a, b and c: computer a can send data to computer b at speed c.

Output

Print one integer: the maximum speed Kotivalo can download data.

Constraints

- $1 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input	Output
4 5 1 2 3 2 4 2 1 3 4 3 4 5 4 1 3	6

Chess inspired problems are a common source of exercises in algorithms classes. Starting with the well known 8-queens problem, several generalizations and variations were made. One of them is the N-rooks problem, which consists of placing N rooks in an N by N chessboard in such a way that they do not attack each other.

Professor Anand presented the N-rooks problem to his students. Since rooks only attack each other when they share a row or column, they soon discovered that the problem can be easily solved by placing the rooks along a main diagonal of the board. So, the professor decided to complicate the problem by adding some pawns to the board. In a board with pawns, two rooks attack each other if and only if they share a row or column and there is no pawn placed between them. Besides, pawns occupy some squares, which gives an additional restriction on which squares the rooks may be placed on.

Given the size of the board and the location of the pawns, tell Professor Anand the maximum number of rooks that can be placed on empty squares such that no two of them attack each other.

Input

The input file contains several test cases, each of them as described below.

The first line contains an integer N ($1 \leq N \leq 100$) representing the number of rows and columns of the board. Each of the next N lines contains a string of N characters. In the i-th of these strings, the j-th character represents the square in the i-th row and j-th column of the board. The character is either '`.`' (dot) or the uppercase letter '`X`', indicating respectively an empty square or a square containing a pawn.

Output

For each test case, output a line with an integer representing the maximum number of rooks that can be placed on the empty squares of the board without attacking each other.

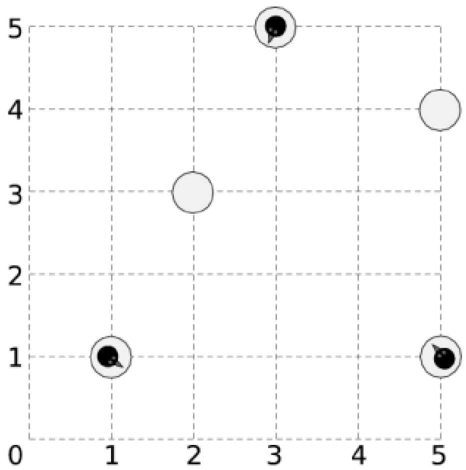
Sample Input

```
5
X....
X....
..X..
.X...
....X
4
....
.X..
....
....
1
X
```

Sample Output

```
7
5
0
```

Somewhere near the south pole, a number of penguins are standing on a number of ice floes. Being social animals, the penguins would like to get together, all on the same floe. The penguins do not want to get wet, so they have use their limited jump distance to get together by jumping from piece to piece. However, temperatures have been high lately, and the floes are showing cracks, and they get damaged further by the force needed to jump to another floe. Fortunately the penguins are real experts on cracking ice floes, and know exactly how many times a penguin can jump off each floe before it disintegrates and disappears. Landing on an ice floe does not damage it. You have to help the penguins find all floes where they can meet.



A sample layout of ice floes with 3 penguins on them.

Input

On the first line one positive number: the number of testcases, at most 100. After that per testcase:

- One line with the integer N ($1 \leq N \leq 100$) and a floating-point number D ($0 \leq D \leq 100000$), denoting the number of ice pieces and the maximum distance a penguin can jump.
- N lines, each line containing x_i, y_i, n_i and m_i, denoting for each ice piece its X and Y coordinate, the number of penguins on it and the maximum number of times a penguin can jump off this piece before it disappears ($-10000 \leq x_i, y_i \leq 10000$, $0 \leq n_i \leq 10$, $1 \leq m_i \leq 200$).

Output

Per testcase:

- One line containing a space-separated list of 0-based indices of the pieces on which all penguins can meet. If no such piece exists, output a line with the single number '-1'.

Sample Input

```
2
5 3.5
1 1 1 1
2 3 0 1
3 5 1 1
5 1 1 1
5 4 0 1
3 1.1
-1 0 5 10
0 0 3 9
2 0 1 1
```

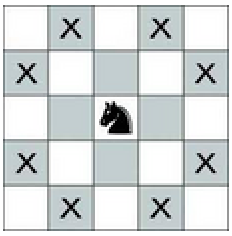
Sample Output

```
1 2 4
-1
```

Problem K. Knights In The Board

Time Limit 1000 ms
Mem Limit 262144 kB
OS Windows

In the world of chess, the knight moves in an L-shape: two squares in one direction and then one square perpendicular, or one square in one direction and then two squares perpendicular. This unique movement allows it to jump over other pieces, reaching eight possible squares from a central position on an empty board.



Possible moves of a knight in a chess board

Given a chessboard of size $N \times N$ and K knights placed on it, determine the minimum number of knights that must be removed from the board so that no two knights are attacking each other. A knight is considered to be attacking another knight if it can move to the square occupied by the second knight in one turn based on its L-shaped movement pattern.

Input

The first line of input contains two integer numbers separated by a space N ($3 \leq N \leq 25$), and K ($1 \leq K \leq N * N$), representing the size of the board, and the number of knights in the board.

Each of the following K lines contains two integer numbers r_i, c_i, representing the i-th knight is placed on row r_i and column c_i in the board. No two knights will be places on the same position.

Output

Output a line with an integer number, the minimum number of knights that should be removed from the board so that no two knights are attacking each other.

Examples

Input	Output
3 4 1 1 1 2 2 1 2 2	0

Input	Output
5 9 3 3 1 2 2 1 1 4 2 4 4 1 5 2 4 5 5 4	2

Input	Output
3 9 1 1 2 3 3 1 3 2 3 3 1 2 1 3 2 1 2 2	4